Shiye (Sally) Cao

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Research Overview

> My research focuses on designing, developing, and deploying AI assistants that empower people in their everyday lives. Taking a **human-centered** approach, I develop theoretical frameworks and computational methods grounded in **empirical studies** of how people assist one another, **participatory design** with end users and stakeholders, and observations of human–AI interaction in user studies and **long-term deployments**.

Keywords: human-computer interaction, embodied AI, AI for education, privacy and security.

Education

PhD Candidate in Computer Science — 2022–present
Johns Hopkins University — Baltimore, MD, USA
Advisors: Chien-Ming Huang and Anqi Liu

M.S.E. in Computer Science — 2021–2022
Johns Hopkins University — Baltimore, MD, USA

B.S. in Computer Science and Applied Mathematics and Statistics — 2018–2021
Johns Hopkins University — Baltimore, MD, USA

Publications

> All publications can be found on my Google Scholar profile. Top-tier conferences (HRI, RSS, CHI, CSCW, IDC) have acceptance rates around 25% and are considered premier venues for Computer Science research.

*Indicates authors contribute equally to the work.

Peer-Reviewed Conference Papers

C.7 **ICMI'26**
ERR@HRI 3.0 Challenge: Multimodal Detection of Errors and Anticipation in Human-Robot Interactions
Maria Teresa Parreira, Micol Spitale, Maia Stiber, **Shiye Cao**, Amama Mahmood, Chien-Ming Huang, Hatice Gunes, and Wendy Ju
Proceedings of the 28th International Conference on Multimodal Interaction
https://arxiv.org/abs/2607.11570

C.6 **IDC'26**
ELLA: Generative AI-Powered Social Robots for Early Language Development at Home
Victor Nikhil Antony*, **Shiye Cao***, Shuning Wang, and Chien-Ming Huang
Proceedings of the 25th Annual ACM Interaction Design and Children Conference
https://dl.acm.org/doi/10.1145/3773077.3806113

C.5 **HRI'26**
Reframing Conversational Design in HRI: Deliberate Design with AI Scaffolds
Shiye Cao, Jiwon Moon, Yifan Xu, Anqi Liu, and Chien-Ming Huang
Proceedings of 2026 ACM/IEEE International Conference on Human-Robot Interaction | Acceptance rate: 23.2%
https://dl.acm.org/doi/10.1145/3757279.3785640

C.4 **ACM-MM'25**
ERR@HRI 2.0 Challenge: Multimodal Detection of Errors and Failures in Human-Robot Conversations
Shiye Cao, Maia Stiber, Amama Mahmood, Maria Teresa Parreira, Wendy Ju, Micol Spitale, Hatice

Gunes, and Chien-Ming Huang
Proceedings of 33rd ACM International Conference on Multimedia
https://dl.acm.org/doi/abs/10.1145/3746027.3762073

C.3 **RSS'25**
Interruption Handling for Conversational Robots
Shiye Cao*, Jiwon Moon*, Amama Mahmood, Victor Nikhil Antony, Ziang Xiao, Anqi Liu, and Chien-Ming Huang
Proceedings of the 2025 Robotics: Science and Systems Conference
https://www.roboticsproceedings.org/rss21/p089.pdf | Acceptance rate: 27.4%

C.2 **CHI'25**
Voice Assistants for Health Self-Management: Designing for and with Older Adults
Amama Mahmood, **Shiye Cao**, Maia Stiber, Victor Nikhil Antony, and Chien-Ming Huang
Proceedings of 2025 ACM Conference on Human Factors in Computing Systems
https://dl.acm.org/doi/10.1145/3706598.3713839 | Acceptance rate: 25.1%

C.1 **HRI'23**
"What If It Is Wrong": Effects of Power Dynamics and Trust Repair Strategy on Trust and Compliance in HRI
Ulas Berk Karli*, **Shiye Cao***, and Chien-Ming Huang
Proceedings of 2023 ACM/IEEE International Conference on Human-Robot Interaction
https://dl.acm.org/doi/10.1145/3568162.3576964 | Acceptance rate: 25.3%

Peer-Reviewed Journal Articles

J.4 **CSCW'24**
Designing for Appropriate Reliance: The Roles of AI Uncertainty Presentation, Initial User Decision, and User Demographics in AI-Assisted Decision-Making
Shiye Cao, Anqi Liu, and Chien-Ming Huang
Proceedings of the ACM on Human-Computer Interaction
Volume 8, Issue CSCW1, Article 41, Pages 1–32
https://dl.acm.org/doi/10.1145/3637318

J.3 **CSCW'23**
How Time Pressure from Different Phases of Decision-Making Influences Human-AI Collaboration
Shiye Cao*, Catalina Gomez*, and Chien-Ming Huang
Proceedings of the ACM on Human-Computer Interaction
Volume 7, Issue CSCW2, Article 277, Pages 1–26
https://dl.acm.org/doi/10.1145/3610068

J.2 **TiiS'23**
Crowdsourcing Thumbnail Captions: Data Collection and Validation
Carlos Aguirre*, **Shiye Cao***, Amama Mahmood, and Chien-Ming Huang
ACM Transactions on Interactive Intelligent Systems
Volume 13, Issue 3, Article 14, Pages 1–28
https://dl.acm.org/doi/10.1145/3589346

J.1 **CSCW'22**
Understanding User Reliance on AI in Assisted Decision-Making
Shiye Cao and Chien-Ming Huang
Proceedings of the ACM on Human-Computer Interaction
Volume 6, Issue CSCW2, Article 471, Pages 1–23
https://dl.acm.org/doi/10.1145/3555572

Refereed Symposium & Workshop Papers

W.4 **UR-RAD'26**
Rapid Conversational Authoring of Interactive Human-Robot Activities
Shiye Cao*, Aureliane Tchamdja*, and Chien-Ming Huang
To appear at the *AAAI Fall Symposium on Unifying Representations for Robot Application Development*

W.3 **HRI'26**
Long-Term In-Home Robotic Support for Children with ASD
Shiye Cao
2026 HRI Pioneers Workshop | Acceptance rate: 23%
https://dl.acm.org/doi/10.1145/3776734.3794629

W.2 **HRI'26**
Practical Trade-offs in Generative Robot Expression Design
Victor Nikhil Antony, **Shiye Cao**, and Chien-Ming Huang
Expressive robot faces with Vizij: a workshop / tutorial hybrid at HRI 2026
https://drive.google.com/file/d/1oZPRrEJ829Ju4gAeAPaSNcZOqEjEAp9b/view

W.1 **TRAIT'23**
Eyes Are the Windows to AI Reliance: Towards Real-Time Human-AI Reliance Assessment
Shiye Cao, Shichang Ke*, Alexandra Mo*, Anqi Liu, and Chien-Ming Huang
2023 CHI Workshop on Trust and Reliance in AI-Assisted Tasks
https://chi-trait.github.io/papers/2023/CHI_TRAIT_2023_Paper_35.pdf

Under-Review Papers

U.3 **Privacy and Security Communication for Robots: Current Practice, Open Challenges, and Recommendations**
Shiye Cao, Laura Fleig, Yaxing Yao, and Chien-Ming Huang

U.2 **Towards Child-Centered Error Management In-the-Wild: Children's Reactions and Recovery Following Robot Errors**
Shiye Cao*, Shuning Wang*, Victor Nikhil Antony, Leyun Kim, and Chien-Ming Huang

U.1 **Sync or Sink: Conversational Pace Matching for Human-Robot Collaboration**
Yifan Yin*, Binze Li*, **Shiye Cao***, Katherine Guerrerio, Seokhyun Baek, Tianmin Shu, and Chien-Ming Huang

Awards

Best Poster Award, Third Annual Celebrating Women in Data Science and AI Symposium 2026
ACM/IEEE HRI Pioneer 2026
CRA Outstanding Undergraduate Researcher Finalist 2022

Teaching Experiences

Teaching Assistant, Introduction to Human-Robot Interaction Spring 2026
Teaching Assistant, Introduction to Human-Computer Interaction Fall 2022
Course Assistant, Machine Learning Fall 2020, Spring 2021, Fall 2021

Professional Service

Organizing Committee: ICMI'26 ERR@HRI 3.0 Challenge, ACM-MM'25 ERR@HRI 2.0 Challenge
Program Committee: 2027 HRI Pioneers Workshop, NeurIPS '22 Gaze Meets ML Workshop
Reviewer: UIST'26 (Outstanding Review), HRI'26, CHI'26 (Outstanding Review), CSCW'25, CHI'25, HRI'25, HRI'24 (Outstanding Review), CSCW'24, CSCW'23, THRI
Student Volunteer: HRI'23, CSCW'23, CHI'23, CSCW'22

Community Outreach

Organized outreach event at the Applied ABC Autism Therapy Center 4/2026
Organized outreach event at the Port Discovery Children's Museum 6/2026
Organized outreach event at the Henderson Hopkins School 8/2026

Research Mentoring

> I have mentored 10+ students on projects leading to publications. Mentees have continued to PhD programs at Yale and the University of Chicago, and one received a CRA Outstanding Undergraduate Researcher Honorable Mention.

PhD Students

Yifan Yin (JHU CS PhD) 2026
Publications: U.1
Binze Li (JHU CS PhD) 2026
Publications: U.1
Laura Fleig (JHU CS PhD) 2026
Publication: U.3

Master's Students

Ulas Berk Karli (JHU Robotics MSE) 2021–2023
Next: CS PhD at Yale University
Publications: C.1
Daniel Han (JHU Applied Mathematics MSE) 2025–2026
Yifan Xu (JHU CS MSE) 2025–2026
Publications: C.5
Shuning Wang (JHU Systems Engineering MSE) 2025–2026
Publications: C.6, U.2 Zeli Ma (JHU CS MSE) 2026

Undergraduate Students

Jiwon Moon (JHU CS BS) 2023–2025
Next: CS PhD at the University of Chicago
2025 CRA Outstanding Undergraduate Researcher Honorable Mention
Publications: C.3, C.5
Yoonjae Kim (JHU Mechanical Engineering BS) 2025–2026
Leyun Kim (JHU CS & Cognitive Science BS) 2026
Publications: U.2
Aureliane Tchamdja (UMBC CS BS, NSF REU) 2026
Publications: W.4

Selected Press

Celebrating women in data science and AI JHU CS News
Aging gracefully with AI JHU CS News
Talking robots learn to manage human interruptions JHU HUB
CS student named finalist for 2022 CRA Outstanding Undergraduate Researcher Award JHU CS News

References

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Anqi (Angie) Liu, Assistant Professor
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Yaxing Yao, Assistant Professor
Department of Computer Science, Johns Hopkins University
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